

Project Design Phase
Proposed Solution Template

Date	30-6-2026
Team ID	23MH1A4257
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional credit card underwriting and approval processes are often slow, manual, and prone to human error or unconscious bias. Financial institutions struggle to quickly process large volumes of applications while accurately assessing credit risk. Poor risk evaluation leads to higher default rates (NPLs), while overly stringent, slow manual checks frustrate creditworthy customers, causing them to abandon the application.
2.	Idea / Solution description	The proposed solution is an automated, Machine Learning-driven Credit Card

		Approval Prediction System. By training predictive models (such as Logistic Regression, Random Forests, or Gradient Boosting techniques) on historical customer financial data—including credit scores, income levels, employment status, age, and existing debt—the system can instantaneously evaluate a new applicant's creditworthiness. It provides a binary classification (Approved/Rejected) alongside a risk probability score to assist financial institutions in making instant, data-backed decisions.
3.	Novelty / Uniqueness	Unlike traditional static credit scoring models (which rely solely on a single rigid FICO score), this solution utilizes advanced ensemble learning capable of capturing complex, non-linear relationships across diverse alternative data points. Furthermore, it incorporates Explainable AI (XAI) frameworks like SHAP or LIME. This ensures the model isn't just a "black box"—it provides transparent reasons for rejection or approval, allowing banks to comply with financial regulations regarding fair lending practices.
4.	Social Impact / Customer Satisfaction	Financial Inclusion: By leveraging machine learning on alternative data, the

		system can assess "credit-invisible" individuals (e.g., young adults or immigrants) who lack a long traditional credit history but are financially responsible. Instant Gratification: Reduces application processing times from days to mere seconds, drastically boosting customer satisfaction and onboarding retention rates. Elimination of Bias: Standardizes the evaluation framework, mitigating potential human bias in the approval process.
5.	Business Model (Revenue Model)	The project can be commercialized using a B2B SaaS (Software-as-a-Service) model targeting fintech startups, regional banks, and credit unions: • Pay-per-API-Call: Charging a micro-fee for every credit application processed through the prediction engine. • Tiered Subscription: Monthly or annual licensing fees based on the volume of applications handled (e.g., Starter, Growth, and Enterprise tiers). • Custom Integration Services: One-time setup fees for integrating the ML pipeline with a bank's

		existing Core Banking Software.
6.	Scalability of the Solution	The solution is highly scalable as it is built on a cloud-native architecture (e.g., AWS or Google Cloud). The prediction engine can be containerized using Docker and orchestrated via Kubernetes, allowing it to automatically scale horizontally to handle thousands of concurrent application requests during peak financial periods without any performance degradation. It can easily adapt to different geographic regions or regulatory markets by retraining the model on local datasets.